Shorten Path

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The problem

- Simplify a Unix path (absolute or relative), resolving "." and ".." and duplicate slashes.
- Leading ".." are kept for relative paths and dropped at the root of absolute paths.

Intuition

- Process path tokens with a stack: dots are ignored, double dots pop, names push.

Approach

- 1. Split on slashes
- 2. Skip empty tokens and single dots
- 3. Double dot pops, or is kept for relative paths
- 4. Join back

Java solution

```java
public class ShortenPath {
    public static String shortenPath(String path) {
        boolean abs = path.startsWith("/");
        ArrayDeque<String> st = new ArrayDeque<>();
        for (String t : path.split("/")) {
            if (t.isEmpty() || t.equals(".")) continue;
            if (t.equals("..")) {
                if (!st.isEmpty() && !st.peekLast().equals("..")) st.pollLast();
                else if (!abs) st.addLast("..");
            } else st.addLast(t);
        }
        String joined = String.join("/", st);
        return abs ? "/" + joined : joined;
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Recap

- Process path tokens with a stack: dots are ignored, double dots pop, names push.
- Complexity: O(n)